1. PROJECT OVERVIEW / BACKGROUND

Introduction

Agriculture plays a major role in the economy of India, and crop pricing directly affects farmers, traders, consumers, and government planning. Market prices of crops often fluctuate because of weather conditions, seasonal demand, transportation issues, storage availability, and supply changes. Predicting future crop prices can help stakeholders make informed decisions regarding selling, purchasing, storage, and investment.

This project focuses on developing a Crop Price Prediction Model using Machine Learning techniques. The system uses historical mandi prices, weather parameters, arrivals data, lag prices, and seasonal indicators to forecast crop prices for the next 7 days and 15 days.

Need of Work / Reason for Selection of this Project

Agriculture is the backbone of the Indian economy, and a large percentage of the population depends on farming for income and livelihood. However, one of the major challenges faced by farmers and traders is the uncertainty of crop market prices. Agricultural commodity prices frequently rise and fall due to several factors such as seasonal production, weather conditions, transportation costs, market demand, government policies, storage availability, and supply chain disruptions. Because of these unpredictable fluctuations, farmers often struggle to decide the right time to sell their produce.

Reasons for Selecting this Project

1. To Help Farmers Maximize Profit by Choosing the Right Selling Time

Farmers usually do not know whether crop prices will rise or fall in the coming days. Because of this uncertainty, they may sell their produce at lower prices and miss better opportunities later. A crop price prediction system can forecast future prices for the next few days or weeks, allowing farmers to

decide the best time to sell their produce. This can directly increase their income and improve financial stability.

2. To Assist Traders in Planning Purchases and Stock Management

Traders buy crops from farmers and sell them in local or larger markets. If prices are expected to rise, traders can purchase and store crops in advance. If prices are expected to fall, they can avoid over-purchasing and reduce losses. Accurate predictions help traders manage stock levels efficiently, improve supply chain operations, and make better business decisions.

3. To Use Artificial Intelligence in the Agriculture Sector

Agriculture in many regions still depends on traditional methods and manual decision-making. By introducing Artificial Intelligence and Machine Learning, the farming sector can become smarter and more efficient. AI models can process large amounts of historical market data, weather data, and seasonal patterns to generate useful insights. This project demonstrates how modern technology can be applied to solve agricultural challenges.

4. To Reduce Financial Losses Caused by Unpredictable Market Fluctuations

Unexpected changes in crop prices can create heavy financial losses for farmers, traders, and suppliers. For example, if a trader purchases stock at a high price and the market price suddenly drops, losses occur. Similarly, farmers lose income when prices fall after harvest. By forecasting possible future trends, stakeholders can prepare in advance and reduce financial risk.

5. To Provide Data-Driven Decision Support

Many agricultural decisions are still based on guesswork, personal experience, or advice from middlemen. A prediction model uses real historical data and statistical patterns to generate reliable forecasts. This helps users make informed decisions instead of relying only on assumptions. Data-driven planning leads to better results and more confidence in decision-making.

6. To Improve Market Transparency

In many rural areas, farmers do not always have access to complete market information. They may not know prices in nearby mandis or expected future trends. A crop price prediction system can improve transparency by giving clear information about likely price movements. This reduces dependence on

intermediaries and empowers farmers with useful knowledge.

Problem Statement

To design and develop a machine learning-based web application that predicts crop prices for the next 7 days and 15 days using historical market data, weather conditions, and crop-related features.

2. METHODOLOGY

Proposed Techniques or Methods

The proposed Crop Price Prediction System is implemented using a combination of modern web technologies and machine learning techniques to ensure accurate predictions and efficient user interaction. The backend of the application is developed using FastAPI, a high-performance Python web framework used for building RESTful APIs. FastAPI is chosen because of its speed, simplicity, automatic documentation support, and ability to handle multiple user requests efficiently. It provides a robust platform for connecting the machine learning model with the frontend application and delivering real-time predictions to users.

For crop price forecasting, the project uses the XGBoost Regression Algorithm, which is one of the most powerful and widely used machine learning algorithms for structured and tabular data. XGBoost (Extreme Gradient Boosting) works by combining multiple decision trees to improve prediction accuracy and reduce errors. It is highly efficient, scalable, and capable of handling complex relationships between input features such as historical crop prices, weather conditions, arrivals data, seasonal indicators, and lag prices. Due to its strong performance and reliability, XGBoost is selected as the core prediction model for this project.

The methodology begins with collecting and preprocessing historical crop market data. Missing values, duplicate records, and inconsistent entries are cleaned to improve data quality. After preprocessing, important features such as previous day prices, moving averages, month, day, season, rainfall, and temperature are prepared as inputs for the model. These features help the algorithm understand market trends and seasonal patterns.

Data Collection

The model uses the following input parameters:

- State
- District
- Mandi
- Crop Name
- Modal Price
- Minimum Price
- Maximum Price
- Arrivals Quantity
- Previous 7-day Price
- Previous 14-day Price
- Previous 30-day Price
- Temperature
- Humidity
- Rainfall
- Day of Year
- Festival Flag
- Harvest Season Flag

Data Preprocessing

Data preprocessing is one of the most important steps in building an accurate and reliable machine learning model. Raw agricultural market data often contains missing values, inconsistent entries, duplicate records, and mixed data formats. If such data is used directly for training, the prediction model may produce poor or inaccurate results.

Preprocessing Steps Performed

1. Handling Missing Values

Real-world mandi and weather datasets often contain missing or incomplete values due to manual entry errors, unavailable records, or system issues. Missing values can negatively affect model performance if left untreated. Therefore:

This ensures that the dataset remains complete and usable for training.

2. Removing Duplicate Records

Sometimes the same crop market entry may appear multiple times in the dataset. Duplicate records can create bias and mislead the model during training. Therefore:

- Repeated rows are detected.
- Duplicate entries are removed carefully.

This improves data consistency and prevents overrepresentation of certain values.

3. Numerical Feature Scaling using Standard Scaler

- Numerical columns are normalized using Standard Scaler.
- This transforms values so that features have a mean of 0 and standard deviation of 1.

Scaling helps the model process features more efficiently and improves stability.

4. Encoding Categorical Features using One-Hot Encoding

Some important dataset columns may contain text categories such as:

- Crop Name
- State Name
- Market Name
- Season
- Variety

Machine learning models cannot directly process text values, so categorical columns are converted into numerical format using One-Hot Encoding. In this

method:

- Each category is converted into a separate binary column (0 or 1).
- Example: Wheat, Rice, Potato become individual columns.

This allows the model to understand categorical information without assuming any ranking between categories.

5. Feature Alignment with Training Columns

- Prediction input features are aligned with saved training columns.
- Missing columns are added with default values (usually 0).
- Extra unnecessary columns are removed.
- Column order is matched exactly with training data.

This ensures smooth and accurate predictions.

6. Data Type Conversion

Sometimes numeric columns are stored as text strings. To avoid errors:

- String values are converted into integers or floating-point numbers.
- Date columns are converted into proper datetime format.

This helps in extracting useful time-based features.

Model Training

Model training is one of the most important stages of the Crop Price Prediction System. In this phase, machine learning algorithms learn patterns, relationships, and trends from historical crop market data so that they can predict future prices accurately. Since the project aims to forecast prices for two different time periods, two separate prediction models are developed and trained independently.

Separate Models Trained

To provide better forecasting for different future durations, two independent XGBoost models are trained:

1. 7-Day Future Price Prediction Model

- Farmers planning immediate selling decisions
- Traders making short-term purchases

- Estimating weekly market price changes
- Managing storage for a few days

This model mainly captures recent historical prices, short-term demand patterns, and immediate weather effects.

2. 15-Day Future Price Prediction Model

This model is trained to predict crop prices after the next 15 days. It focuses on medium-term trends and slightly broader market behavior. It is useful for:

- Farmers deciding whether to store produce longer
- Traders planning bulk purchases
- Anticipating upcoming seasonal demand changes
- Better stock and supply chain planning

Training Process

1. Dataset Preparation

2. Train-Test Split

3. Model Learning using XGBoost

4. Hyperparameter Tuning

Backend Development

The backend of the Crop Price Prediction System is developed using FastAPI, a modern and high-performance Python web framework used for building APIs quickly and efficiently. FastAPI is selected because it provides fast execution speed, automatic API documentation, easy integration with machine learning models, and excellent support for asynchronous operations. It acts as the bridge between the frontend user interface and the trained machine learning models.

Why FastAPI is Used

FastAPI offers several advantages for this project:

- High performance similar to Node.js and Go
- Easy development with Python syntax
- Automatic Swagger API documentation
- Input validation using Pydantic models

Because of these features, FastAPI is ideal for machine learning-based web applications.

REST APIs Developed

The backend exposes RESTful APIs that can be accessed by frontend applications or external systems.

1. /predict → Predict Oilseed Crop Prices

This API endpoint is designed to predict prices of oilseed-related crops such as:

- Mustard
- Soybean
- Groundnut
- Sunflower
- Sesame
- Other edible oil crops

Working Process:

- User sends crop-related input data
- API validates input fields
- Preprocessing steps are applied
- Trained XGBoost model is loaded
- Future prices for 7 days and 15 days are predicted
- Response is returned in JSON format

Example Output:

{

"crop": "Mustard",

"predicted_price_7_day": 5820,

"predicted_price_15_day": 6015

}

Benefits:

- Helps farmers know future oilseed crop prices
- Supports better selling decisions
- Useful for edible oil market planning

2. /predict_trade_price → Predict Trader Market Prices.

Working Process:

- Accepts trading market parameters
- Uses market trends and historical price data
- Predicts expected future trade prices
- Returns price insights for buying/selling decisions

Example Output:

{

"predicted_trade_price_7_day": 6150,

"predicted_trade_price_15_day": 6280

}

Backend Functional Workflow

The backend performs the following operations:

1. Input Data Validation

Using Pydantic schemas, FastAPI checks:

- Required fields present or not
- Correct data types
- Valid numeric ranges
- Missing values handling

2. Model Loading

Saved model files are loaded using:

- joblib
- pickle

This allows quick prediction without retraining.

Deployment Ready Features

- CORS enabled for frontend integration
- JSON request-response structure
- Error handling with HTTP exceptions

3. PROJECT REQUIREMENTS

The requirements are divided into three main categories:

1. Software Requirements
2. Hardware Requirements
3. Other Resources / Data Requirements

3.1 Software Requirements

The following software tools, frameworks, and libraries are used in the development of this project:

1. Python 3.x

Python is the main programming language used for this project. It is selected because of its simplicity, rich ecosystem, and strong support for machine learning, data science, and backend development.

2. FastAPI

FastAPI is used to build the backend REST APIs. It provides high performance, automatic documentation, and easy integration with machine learning models.

3. Uvicorn Server

Uvicorn is an ASGI server used to run FastAPI applications efficiently.

4. Pandas

Pandas is a Python library used for handling tabular data and datasets.

5. NumPy

NumPy is used for numerical operations and array processing.

6. Scikit-learn

Scikit-learn is used for preprocessing, train-test split, scaling, and model evaluation.

7. XGBoost

XGBoost is the core machine learning algorithm used for crop price prediction.

8. Joblib

Joblib is used to save and load trained machine learning models.

9. VS Code / PyCharm

These are development environments used for writing and managing code.

10. Postman

Postman is used for testing APIs before frontend integration.

11. React.js / Vite (Optional Frontend)

3.2 Hardware Requirements

Hardware resources are required for development, model training, and application deployment.

Minimum Requirements

These specifications are sufficient for development and small-scale testing:

Processor:

- Intel i3 or equivalent dual-core processor

RAM:

- Minimum 4 GB RAM

Storage:

- At least 20 GB free disk space

Internet:

- Basic internet connection for package installation and deployment

Recommended Requirements

These specifications are recommended for better performance and faster training:

Processor:

- Intel i5 / Intel i7 / AMD Ryzen 5 or above

RAM:

- 8 GB RAM or higher

Storage:

- SSD storage for faster read/write operations

Internet:

- Stable high-speed internet connection

Additional:

- Dedicated GPU (optional for larger future models)

Other Resources

- Historical mandi crop price datasets
- Weather datasets (temperature, humidity, rainfall)
- Agricultural seasonal calendar
- Research papers on price forecasting
- Cloud platform for deployment (Render / AWS / Railway)

4. REFERENCES

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